

ALOFI, Niue

WMO: 918220

Lat: 19.067S

Lon: 169.917W

Elev: 22

StdP: 101.06

Time Zone: -11.00 (W11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	17.2	13.0	9.4	19.8	14.0	10.0	20.4	11.0	23.0	10.0	23.2	0.7	150	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	30.8	26.0	30.2	25.8	29.9	25.7	27.0	29.1	26.7	28.8	26.4	28.6	4.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	28.2	26.1	21.5	28.1	25.9	21.3	27.9	85.3	29.6	83.9	29.1	82.4	28.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.6	8.6	7.8	14.4	31.6	1.7	0.5	13.1	32.0	12.1	32.3	11.2	32.5	9.9	32.9		
			13.0	28.0	1.3	0.7	12.1	28.5	11.3	28.9	10.6	29.3	9.6	29.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.8	26.3	26.7	26.5	25.8	24.5	23.5	23.0	22.8	23.5	24.1	25.0	25.9	(6)
	DBStd	1.88	1.07	0.93	1.05	0.99	1.17	1.51	1.79	1.56	1.38	1.28	1.37	1.16	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	5397	505	469	511	474	450	406	403	397	404	437	449	492	(10)
	CDD18.3	2355	247	236	253	224	192	156	144	138	154	179	199	234	(11)
	CDH23.3	16273	2139	2177	2257	1733	1127	729	613	561	678	987	1368	1904	(12)
	CDH26.7	3156	504	558	620	370	127	67	41	46	57	130	243	394	(13)
(14) Wind	WSAvg	3.4	3.4	3.4	2.7	3.1	3.4	3.3	3.0	3.7	4.0	3.8	3.7	3.5	(14)
(15) Precipitation	PrecAvg	2083	288	218	275	239	167	110	93	105	119	134	155	180	(15)
	PrecMax	3184	973	511	898	495	435	536	271	365	361	344	366	582	(16)
	PrecMin	839	61	15	25	7	8	14	8	6	15	7	4	22	(17)
	PrecStd	538	203	128	183	127	101	108	71	101	95	91	92	142	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	31.2	31.2	30.5	29.2	29.1	28.7	28.8	28.9	29.5	30.2	30.8	(19)
		MCWB	26.2	26.0	26.3	25.6	25.6	25.1	24.5	24.3	24.5	24.6	25.3	25.5	(20)
	2%	DB	30.2	30.6	30.5	30.0	28.7	28.1	27.8	28.0	28.0	28.9	29.2	30.0	(21)
		MCWB	25.9	25.9	26.0	25.5	25.1	24.6	24.1	23.8	24.1	24.5	25.0	25.4	(22)
	5%	DB	29.9	30.1	30.1	29.2	28.0	27.4	27.0	27.1	27.2	28.1	28.9	29.2	(23)
		MCWB	25.7	25.8	25.9	25.2	24.5	24.0	23.5	23.2	23.4	23.9	24.8	25.1	(24)
	10%	DB	29.1	29.6	29.7	28.9	27.2	26.8	26.1	26.2	26.5	27.2	28.1	28.8	(25)
		MCWB	25.2	25.6	25.7	25.0	23.7	23.6	22.8	22.6	23.0	23.4	24.4	24.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.5	27.5	27.2	26.7	25.9	25.9	25.6	25.7	26.1	26.6	26.7	(27)
		MCDB	29.3	29.3	29.5	28.9	28.2	27.9	27.2	27.3	27.6	28.0	28.4	28.7	(28)
	2%	WB	26.7	26.8	26.9	26.6	26.0	25.2	25.0	24.7	25.0	25.5	26.1	26.3	(29)
		MCDB	28.9	29.0	29.2	28.5	27.6	27.2	26.8	26.7	26.8	27.6	28.2	28.5	(30)
	5%	WB	26.3	26.5	26.5	26.1	25.5	24.7	24.4	24.1	24.4	24.9	25.6	25.9	(31)
		MCDB	28.5	28.7	28.9	28.2	27.1	26.7	26.2	26.3	26.4	27.2	27.8	28.3	(32)
	10%	WB	26.0	26.1	26.2	25.7	24.8	24.3	23.8	23.5	23.8	24.3	25.1	25.6	(33)
		MCDB	28.2	28.5	28.6	27.8	26.6	26.1	25.5	25.6	25.7	26.4	27.4	27.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	6.3	6.0	5.9	6.0	6.2	6.3	5.9	6.2	6.1	5.7	(35)
		MCDBR	6.4	6.4	6.9	6.8	6.7	6.9	6.8	7.6	6.9	7.3	6.8	6.4	(36)
	5% WB	MCWBR	3.2	3.1	3.6	3.9	4.0	4.1	4.1	4.5	3.9	3.9	3.5	3.3	(37)
		MCWBR	5.5	5.6	6.1	6.0	6.1	5.9	5.8	6.5	6.1	6.4	6.4	5.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.386	0.383	0.377	0.386	0.380	0.378	0.368	0.374	0.395	0.383	0.380	0.384	(40)
		taud	2.623	2.631	2.663	2.622	2.617	2.601	2.624	2.602	2.543	2.581	2.613	2.607	(41)
	Ebn at Noon		959	950	934	887	855	837	859	886	902	939	957	961	(42)
		Edh at Noon	102	100	94	91	86	84	84	92	104	104	103	104	(43)
(44) All-Sky Solar Radiation	RadAvg		6.17	6.06	5.63	4.87	4.09	3.79	4.03	4.69	5.33	5.95	6.16	6.40	(44)
	RadStd		0.63	0.60	0.44	0.36	0.32	0.26	0.19	0.28	0.32	0.43	0.42	0.44	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page